

DIGITAL SOVEREIGNTY IN THE EUROPEAN UNION: FIVE CHALLENGES FROM A NORMATIVE PERSPECTIVE

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Abstract

This paper aims to make sense of *digital sovereignty* in the European Union by examining both its meanings and its obstacles. I start by unpacking the term through a brief exploration of its use in European discourse, and reflect on two of its dimensions that are prevalently highlighted: *economic* and *normative* digital sovereignty (section 2). Focusing primarily on the latter, I subsequently examine five challenges that European digital sovereignty currently faces, and that must be tackled if the EU wishes to maintain its ability to exercise its sovereign powers. These include the significant role of private actors in the public digital sphere, interference by foreign actors, Europe's rule of law crisis, the risk of poor digital governance, and the lack of digital literacy (section 3). Finally, I offer some concluding remarks, acknowledging the difficult balances the EU will need to strike in its digital policies (section 4).

1 Introduction

In recent years, the term *digital sovereignty* increasingly found its way into the European Union's political discourse. As with many vessels of public speech, the fact that this notion has been used to denote different things by different people did not diminish its proliferation. But what precisely does it signify? From a legal perspective, associating the words 'sovereignty' with 'Europe' traditionally evokes the idea of the autonomy of the EU legal order. Indeed, as far back as *Costa v E.N.E.L.*, the Union has been conceptualized by the Court of Justice of the European Union (CJEU) as an independent source of law,¹ whereby EU law

¹ Case 6/64, Judgment of the Court of 15 July 1964, *Flaminio Costa v E.N.E.L.*, ECLI:EU:C:1964:66: "By creating a Community of unlimited duration, having its own institutions, its own personality, its own legal capacity and capacity of representation on the international plane and, more particularly, real powers stemming from a limitation of sovereignty or a transfer of powers from the States to the Community, the Member States have limited their sovereign rights, albeit within limited fields, and have thus created a body of law which binds both their nationals and themselves."

presents a “*legal order that has the capacity to operate as a self-referential system of norms that is both coherent and complete*”.²

European sovereignty hence implies a vision of the EU as an autonomous agent.³ This holds true when considering the EU as *legislating* (having the ability to adopt legislation, through the European Parliament and Council), as *adjudicating* (having the ability to engage in judicial decision-making, through the CJEU alongside national courts), as *regulating* (having the ability to regulate, through the European Commission and regulatory agencies) and as *enforcing* (having the ability to enforce EU law, whether through the European Commission or a national authority). Arguably, the fact that the EU legal order co-exists with the legal orders of its Member States complicates yet does not diminish this picture – it merely implies a delimitation of competence and jurisdiction.⁴ While sovereignty is traditionally conceptualized at the level of nation-states, it has been amply applied to the European context as well, to the extent and degree that certain competences have been allocated to the EU by its Member States. Nor is sovereignty undermined by the fact the EU can only operate legitimately within the boundaries stipulated by human rights, democracy and the rule of law. Instead, these boundaries signify a conception of sovereignty linked to the Kantian idea of self-governance based on reason, and thus to deliberately choose the limits within which one operates.⁵ And even though some scholars claimed that the concept of sovereignty has become moot in light of globalization⁶ or should even be buried altogether,⁷ it has also been acknowledged that “*today, in contrast, it seems that there is a move towards increasing sovereignty*”,⁸ and that it can be seen as an elastic concept with different shades and shapes.

² Koen Lenaerts, José A Gutiérrez-Fons and Stanislas Adam, ‘Exploring the Autonomy of the European Union Legal Order’ (2021) 81 *Zeitschrift für ausländisches öffentliches Recht und Völkerrecht* / Heidelberg Journal of International Law 47, 48.

³ On the relationship between sovereignty and autonomy, see e.g. Raf Geenens, ‘Sovereignty as Autonomy’ (2017) 36 *Law and Philosophy* 495.

⁴ See e.g. Andrew Moravcsik, *The Choice for Europe: Social Purpose and State Power from Messina to Maastricht*. (UCL press 1999).

⁵ See Immanuel Kant, *Groundwork for the Metaphysics of Morals* (Lara Denis ed, Thomas Kingsmill Abbott tr, Broadview 2005). See also Paul Guyer, ‘Kant on the Theory and Practice of Autonomy’ in Ellen Frankel Paul, Jr Miller Fred D and Jeffrey Paul (eds), *Autonomy*, vol 20 (Cambridge University Press 2003).

⁶ A contemporary critique of sovereignty – sometimes also formulated as a threat – arises from the context of globalization, which seems to undermine the Westphalian idea of the authority of the nation state, and instead points an erosion or at least multi-levelling of sovereignty. The ‘pooling’ of sovereignty by national states through the integration project of the EU is sometimes seen as an answer to this threat. While I will not discuss this critique here, the interested reader may refer to e.g. Richard Bellamy, ‘Sovereignty, Post-Sovereignty and Pre-Sovereignty: Three Models of the State, Democracy and Rights within the EU’ in Neil Walker (ed), *Sovereignty in Transition* (Hart Publishing 2003); Nathalie Brack, Ramona Coman and Amandine Crespy, ‘Sovereignty conflicts in the European Union’ (2019) 4 *Les Cahiers du Cevipol* 3.

⁷ See e.g. Don Herzog, *Sovereignty, RIP* (Yale University Press 2020).

⁸ Roxana Vatanparast, ‘Data Governance and the Elasticity of Sovereignty’ (2020) 46 *Brooklyn Journal of International Law* 1, 3.

In this paper, it is not my intention to revisit the origins and meanings of sovereignty, nor do I want to delve into the question of just how much sovereignty a supranational entity like the EU can claim. Instead, I want to examine what we are to make of sovereignty in Europe when it is preceded by the prefix ‘*digital*’. One could understand the notion of *digital sovereignty* as a sub-category of the EU’s sovereignty; a sub-category that specifically pertains to its sovereign powers vis à vis digital infrastructures. Simultaneously, the deployment of this notion seems to push our conception of sovereignty outwards, beyond its traditional features, towards a reality that forces the EU to reckon with new market structures and security threats induced by the disruptive nature of digital technologies.

The digital realm hence alters and at the same time challenges the traditional notion of sovereignty, leading the ‘prefix’ to do more work than merely expressing a sub-category thereof. This is because digital technologies are profoundly reshaping both societies and individuals, including how we act, communicate, and think. Inevitably, this influence also extends to political institutions, and thus to the very ability to exercise sovereignty in the first place. The recent proliferation of new EU legislation in the digital sphere – from the Digital Services Package⁹ and the AI Package¹⁰ to the Data Package¹¹ – can hence be seen not only as an exercise of EU sovereignty to ensure the protection of EU values (from fair competition to democracy), but also as an attempt to maintain and protect its sovereignty in an evolving world. Given society’s ever-increasing reliance on digital technology, one could even ask whether the concept of ‘sovereignty’ still has any meaning without its digital prefix.

Against this background, this paper aims to make sense of *digital sovereignty* in the European Union by examining both its meanings and its obstacles. I start by unpacking the term through a brief exploration of its use in European discourse, and reflect on two of its dimensions that are prevalently highlighted: *economic* and *normative* digital sovereignty (section 2). Focusing primarily on the latter, I subsequently examine five challenges that European digital sovereignty currently faces, and that must be tackled if the EU wishes to maintain its ability to exercise its sovereign powers (section 3). These include the significant role of private actors in the public digital sphere, interference by foreign actors, Europe’s rule of law crisis, the risk of poor digital governance, and the lack of digital literacy. Finally, I offer some concluding remarks, acknowledging the difficult balances the EU will need to strike in its digital policies (section 4).

⁹ This includes the Digital Markets Act and the Digital Services Act, both of which have been adopted in 2022.

¹⁰ This includes the AI Act and the AI Liability Directive, both of which are still being negotiated in 2023.

¹¹ This includes the Data Governance Act, adopted in 2022, and the Data Act, on which political agreement was reached between the Council and Parliament in 2023.

2 Unpacking digital sovereignty in the EU

There is no single definition of sovereignty. Like many value-laden political concepts, it has in the past been characterized as a so-called essentially contested concept.¹² This is not different when it comes to *digital* sovereignty, and in this paper I will hence not make an attempt at defining this notion. Instead, I will briefly explore in this section how this notion is currently being used in the EU context, with the aim of unpacking the core elements of its meaning for the purpose of our analysis.

2.1 Exploring digital sovereignty

Though it already entered public discourse before, the year 2020 seemed to have marked a high point for the term digital sovereignty. It was the year that French President Macron most expressly proclaimed the need to ensure digital sovereignty at European level. According to him, this implies that it is “*up to Europe to define the regulatory framework that it imposes upon itself, as it entails both the protection of individual liberties and the economic data of our companies, laying at the heart of our sovereignty and our concrete operational capacity to act in an autonomous manner.*”¹³ 2020 was also the year that Germany explicitly put forward digital sovereignty as a *leitmotiv* of European digital policy during its presidency of the European Council, “*to ensure that, in the context of fair competition, we increase our prosperity, protect our security and uphold our values.*”¹⁴ And it was the year that Thierry Breton, European Commissioner for Internal Market, announced several initiatives to strengthen European digital sovereignty, which he equated with “*Europe defending its strategic interests*” and “*making sure that anyone who invests, operates and bids in Europe respects our rules and values.*”¹⁵

The fact that the calls for digital sovereignty only intensified thereafter is no surprise. The COVID-19 virus, which the World Health Organization formally started calling a pandemic in March 2020, offered a grim reminder of our collective dependence on digital technologies. Never was this dependency more felt than the

¹² Georg Sørensen, ‘Sovereignty: Change and Continuity in a Fundamental Institution’ (1999) 47 Political Studies 590; Dan Sarooshi, ‘The Essentially Contested Nature of the Concept of Sovereignty: Implications for the Exercise by International Organizations of Delegated Powers of Government’ (2004) 25 Michigan Journal of International Law 1107.

¹³ Author’s translation from French. See Emmanuel Macron, ‘Discours du Président Emmanuel Macron sur la stratégie de défense et de dissuasion devant les stagiaires de la 27ème promotion de l’école de guerre’ (7 February 2020) <<https://www.elysee.fr/emmanuel-macron/2020/02/07/discours-du-president-emmanuel-macron-sur-la-strategie-de-defense-et-de-dissuasion-devant-les-stagiaires-de-la-27eme-promotion-de-lecole-de-guerre>> accessed 1 June 2023.

¹⁴ Germany’s Presidency of the Council of the European Union, ‘Together for Europe’s Recovery - Programme for Germany’s Presidency of the Council of the European Union’ (Federal Foreign Office 2020) 8.

¹⁵ Thierry Breton, ‘Speech by Commissioner Thierry Breton at Hannover Messe’ (Hannover Messe Digital Days, 20 July 2020) <https://ec.europa.eu/commission/presscorner/detail/es/speech_20_1362> accessed 1 June 2023.

moment when the world practically stopped, and digital technology became our only means of communicating, not only for private, but also for professional and public ends. More importantly, however, it also underlined our dependency on the enablers and providers of digital technology and its infrastructure. Consider, for instance, the Google-Apple partnership to develop application programming interfaces and operating system-level technology to allow for Bluetooth-based contact tracing. Rapidly, governments across the world started relying on this infrastructure to manage the pandemic – and to indirectly manage the civil liberties of their citizens. Yet as Luciano Floridi points out, the same companies saw no issue in admitting that they also had the power to “*disable the system on a regional basis when no longer needed*”.¹⁶

Evidently, this reliance on digital technology, and on the entities providing it, raised new questions in relation to state sovereignty – not only at the national but also at the Union level. In her 2020 State of the Union address, Commission President Ursula von der Leyen therefore acknowledged the need to “*double down*” investments in “*European tech sovereignty*” so as to “*shape our digital transformation according to our own rules and values*”. Concretizing this ambition, in early 2021 the European Commission published a Communication titled *2030 Digital Compass: the European way for the Digital Decade*. The Communication stressed how the pandemic “*exposed the vulnerabilities of our digital space*” and “*its dependencies on non-European technologies*.”¹⁷ The Communication also set out “*concrete targets*” to ensure that Europe’s digital ambitions would be met, hence reflecting the desire of the EU and its Member States to take (back) control over Europe’s digital future.¹⁸ Indeed, the Union is not entirely helpless when it comes to protecting its digital sovereignty, and EU institutions have been exploring several ways to address those new sovereignty-related questions. In this regard, Floridi also points out that “*between companies and states, the former can determine the nature and speed of change, but the latter can control the direction of change*”.¹⁹ Indeed, companies may be the ones developing the technology, yet states are able to draw on their sovereignty to set the boundaries within which such development ought to take place.

The ability to decide on this direction is also echoed in Larsen’s use of the term digital sovereignty, which according to him “*refers to a nation’s ability to control its digital destiny*”, while adding that this “*may include control over the entire AI supply*

¹⁶ Luciano Floridi, ‘The Fight for Digital Sovereignty: What It Is, and Why It Matters, Especially for the EU’ (2020) 33 *Philosophy & Technology* 369, 369.

¹⁷ European Commission, ‘2030 Digital Compass: The European Way for the Digital Decade’ (2021) Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of Regions COM/2021/118 final.

¹⁸ *Ibid.*

¹⁹ Floridi (n 16) 371.

chain, from data to hardware and software”.²⁰ Furthermore, some references to digital sovereignty go beyond the state and explicitly include the role of natural and legal persons, either directly, e.g. avowing that citizens and businesses should be able to independently assert control over their relationship with digital technology or indirectly, e.g. avowing that states must ensure to “*protect their citizens and businesses from the manifold challenges to self-determination in the digital sphere*”.²¹

Lifting this to the European context, European digital sovereignty can thus be understood as the ability of the European Union – including its citizens and businesses – to autonomously decide on their relationship with (providers of) digital technology.²² This implies the exercise of control over two entwined elements, namely (a) the direction or ‘destiny’ of digital technology within the EU, and (b) the concrete digital infrastructure that enables the technology. Let me briefly discuss each of these in turn, starting with the latter.

2.2 Economic Digital Sovereignty

Many digital sovereignty claims pertain to control over digital infrastructure, which is typically associated with productive capabilities and economic power in the digital sphere. Let us call this *economic digital sovereignty (EDS)*, or the ability of the EU to control the production factors of digital technologies and their underlying infrastructure. It is in this context that the EU’s dependency on (foreign) technology actors is often problematized. Consider, for instance, the 2023 Opinion of the European Economic and Social Committee on Digital Sovereignty, which in its first paragraph states that “*despite significant progress to enhance the EU’s digital sovereignty, there is still heavy reliance on non-EU-based tech companies. This is limiting the EU’s leadership and strategic autonomy in the digital world, and in turn limiting the EU’s economic growth potential*”.²³ Think also of Commissioner Breton’s abovementioned speech, in which he stated that: “*to be ahead of the curve, we need to develop suitable European infrastructures allowing the storage, the use, and the creation of data-based applications or Artificial Intelligence services. I consider this as a major issue of Europe’s digital sovereignty*”.²⁴ In other words, the EU must limit its dependency on and the influence of

²⁰ Benjamin Cedric Larsen, ‘The Geopolitics of AI and the Rise of Digital Sovereignty’ (*Brookings*, 8 December 2022) <<https://www.brookings.edu/research/the-geopolitics-of-ai-and-the-rise-of-digital-sovereignty/>> accessed 8 April 2023.

²¹ Julia Pohle and Thorsten Thiel, ‘Digital Sovereignty’ (2020) 9 Internet Policy Review. Having analyzed several digital sovereignty claims, Pohl and Thiel suggest these can be classified based on whether they seek to enhance self-determination by states, companies or individuals.

²² The terms digital sovereignty, data sovereignty and technological sovereignty are sometimes used distinctly (see e.g. Figure 1, page 2 of Annegret Bendiek and Isabella Stürzer, ‘Advancing European Internal and External Digital Sovereignty. The Brussels Effect and the EU-US Trade and Technology Council’ (2022) 20 SWP Comment 1.), yet I use the first term as also encompassing the two latter.

²³ Opinion of the European Economic and Social Committee on Digital Sovereignty: a crucial pillar for EU’s digitalisation and growth 2023 (2023/C 75/02).

²⁴ Breton (n 15).

technology-actors in the US and China and bolster its own means of technology production. Concerns in this regard range from Europe's economic competitiveness and security, and its geopolitical position, to the general welfare of its constituents.

As a result, the EU has been drawing on areas in which its legal competences are strongest – from internal market law to competition law – to take measures aimed at enhancing its economic digital sovereignty.²⁵ The European Chips Act is a case in point.²⁶ This regulation, proposed by the Commission in February 2022 and politically agreed upon by the Parliament and the Council in April 2023, sets up the *Chips for Europe Initiative* to reinforce the Union's competitiveness, resilience and innovation capacity in the semiconductor industry. Concretely, it establishes a framework to financially support manufacturing facilities to ensure security of supply, and it creates a monitoring mechanism for the semiconductor value chain to respond to disruptions of the supply of semiconductors that have an impact on the proper functioning of the internal market.²⁷ Other examples are EU initiatives around data and data governance,²⁸ network connectivity,²⁹ edge and cloud computing³⁰ and cybersecurity.³¹

Importantly, in the context of these initiatives, the EU also clearly acknowledged their importance beyond the economy, and has expressed their link with core EU values, such as fundamental rights and democracy. By gaining more control over the production of digital technology and its infrastructure, the Union aims to increase its grip over the normative direction that digital technology is taking. This direction is also clearly expressed in the Parliament and Council's Decision on the Digital Decade Policy Programme 2030, which expresses as one of its primary objectives in Article 3: “*promoting a human-centred, fundamental-rights-based, inclusive, transparent and open digital environment where secure and interoperable digital technologies and services observe and enhance Union principles, rights and values and are accessible*

²⁵ See also Dennis Broeders, Fabio Cristiano and Monica Kaminska, ‘In Search of Digital Sovereignty and Strategic Autonomy: Normative Power Europe to the Test of Its Geopolitical Ambitions’ [2023] *Journal of Common Market Studies* 1, 5.

²⁶ Proposal for a Regulation of the European Parliament and of the Council establishing a framework of measures for strengthening Europe's semiconductor ecosystem (Chips Act) 2022 (COM(2022) 46 final).

²⁷ The legal basis of the EU's Chips Act rests on two sets of provisions, namely Articles 173(3), 182(1) and 183 TFEU as regards the regulation's goal to accelerate the semiconductor production of the Union's industry, and Article 114 TFEU as regards its goal to create a harmonised legal framework for increasing the Union's resilience and security of supply.

²⁸ For instance the European Data Strategy of February 2020, which formed the basis of initiatives such as the Data Governance Act (Regulation 2022/868 of 30 May 2022) and the Data Act (still under negotiations after a Commission proposal from February 2022). To a certain extent, the General Data Protection Regulation also fits in this list, as it not only sets limitations but also seeks to enable the flow of personal data.

²⁹ For instance the Gigabit Infrastructure Act, proposed by the Commission in February 2023.

³⁰ For instance the launch of the European Alliance for Industrial Data, Edge and Cloud in July 2021, which builds on the European data strategy and the Declaration on European Cloud signed by EU Member States in October 2020.

³¹ For instance the Cybersecurity Act (Regulation 2019/881 of 17 April 2019).

to all, everywhere in the Union”. The economic is hence inherently entwined with the normative.

2.3 Normative digital sovereignty

As noted above, claims about the EU’s digital sovereignty not only extend to controlling digital technology and its infrastructure as such (for instance to reduce Europe’s dependence on foreign providers thereof), but also allude to the self-determination of Europe’s digital ‘destiny’ through norm-setting. By expressing the values that digital technology providers should abide by and the conditions to which technology should be subjected, Europe can shape how technology will shape its society. This is hence often associated with legal powers and competences, or the capacity to autonomously set rules on digital technology and its providers. Let us call this *normative digital sovereignty (NDS)*, or the ability of the EU to control the normative framework within which digital technologies operate in Europe. As noted above, within the EU legal order, this framework can be shaped through legislative, adjudicative, regulatory and enforcement actions. Importantly, however, it encompasses an array of normative instruments going well beyond mere binding legislation, also including soft law initiatives, for instance.

Accordingly, in addition to relying on its economic power, Europe can adopt norms or requirements for the technology’s design, provision and use. To the extent those norms also affect technology providers that are established outside the EU, Europe can seek to impose its normative framework beyond its own constituents, and hence try to ensure that the digital technology it is dependent on – even if foreign – nevertheless complies with its values. This has influentially been coined by Anu Bradford as the so-called Brussels effect.³² Many of the EU initiatives mentioned above are in fact hybrid in scope.³³ While they may focus primarily on norm-setting, they typically also have a strong economic undertone, given that the legal basis typically concerns Article 114 TFEU’s internal market harmonization.

The Commission’s proposal for an AI Act is a good example thereof.³⁴ This upcoming regulation establishes a harmonized legal framework for AI systems that are placed on the Union market to ensure they “*are safe and respect existing law on fundamental rights and Union values*”.³⁵ Building on the *Ethics Guidelines for Trustworthy*

³² Anu Bradford, *The Brussels Effect: How the European Union Rules the World*. (Oxford University Press 2020).

³³ Broeders, Cristiano and Kaminska (n 25) 9.

³⁴ European Commission, Proposal for a Regulation of the European Parliament and the Council laying down harmonised rules on Artificial Intelligence (Artificial Intelligence Act) and amending certain Union legislative acts. 2021 [COM(2021) 206 final]. The AI Act relies on Article 114 TFEU as its legal basis and, more limitedly, Article 16 TFEU as concerns the specific rules it contains around the processing of personal (biometric) data.

³⁵ Ibid, Explanatory Memorandum.

AI developed by the Commission's High-Level Expert Group on AI in 2019,³⁶ the AI Act's primary aim is to set out a vision for Europe's AI destiny, ensuring that it is compatible with respect for fundamental rights, democracy and the rule of law by imposing a set of requirements on the provision and use of such systems. Evidently, this aim is not an ideal match with the product-safety regulation approach that the Commission has proposed to achieve that aim (and, unsurprisingly, the proposal has not been spared from criticism).³⁷ That said, the AI Act can also be seen as a prime example of Europe making use of its regulatory prowess to impose its vision through a comprehensive set of rules that will have extra-EU effects, thus potentially gaining a first-mover competitive advantage in the race to AI regulation.³⁸ Moreover, at the same time, the Commission has been careful to stress that the enhanced legal certainty which the AI Act will generate shall also benefit the economic development and uptake of AI in Europe. The normative direction-setting aims may well be at the forefront of the regulation, yet one of its explicit objectives also concerns the facilitation of investment and innovation in AI, for instance through the establishment of regulatory sandboxes in all Member States.

Economic and normative digital sovereignty can hence be seen as two sides of the same coin.³⁹ Critics have sometimes even blamed the EU for relying on its legislative powers to compensate for its lack of economic power in the digital sphere, with China and the US being portrayed as *producing* technology, and Europe merely *regulating* it. Yet regardless of Europe's digital sovereignty standing, its constitutional obligation to adhere to human rights, democracy and the rule of law also means it must respect and protect these values across its digital policies. In other words, the Union's objectives to expand its economic digital sovereignty must not stand in the way of – but should rather be conducive to – the protection of the EU's core values as listed in Article 2 TEU.⁴⁰

³⁶ Nathalie A Smuha, 'The EU Approach to Ethics Guidelines for Trustworthy Artificial Intelligence' (2019) 20 Computer Law Review International 97.

³⁷ See e.g. Nathalie A Smuha and others, 'How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act' (Social Science Research Network 2021) <<https://papers.ssrn.com/abstract=3899991>>; Michael Veale and Frederik Zuiderveen Borgesius, 'Demystifying the Draft EU Artificial Intelligence Act' (2021) 22 Computer Law Review International 97; Nathalie A Smuha, 'Beyond the Individual: Governing AI's Societal Harm' (2021) 10 Internet Policy Review; Jeremias Adams-Prassl, 'Regulating Algorithms at Work: Lessons for a "European Approach to Artificial Intelligence"' (2022) 13 European Labour Law Journal 30.

³⁸ Nathalie A Smuha, 'From a "Race to AI" to a "Race to AI Regulation": Regulatory Competition for Artificial Intelligence' (2021) 13 Law, Innovation and Technology 57.

³⁹ One could also perceive the economic dimension as normative, since economic policies are inevitably underpinned by an implicit or explicit normative vision of society too.

⁴⁰ This appears to be explicitly acknowledged in the *European Declaration on Digital Rights and Principles for the Digital Decade*, signed by the Parliament and Council in December 2022. See The European Parliament, the Council and the Commission, 'European Declaration on Digital Rights and Principles for the Digital Decade' (2022) <<https://digital-strategy.ec.europa.eu/en/library/european-declaration-digital-rights-and-principles>>. Recital 6 of the Declaration's Preamble states that "EU way for the digital transformation of our societies and

3 Five challenges for European Digital Sovereignty

In the sections above, I conceptualized European digital sovereignty as the EU's ability to autonomously decide on its relationship with digital technology, both economically and normatively. This implies the ability to control the production factors of digital technologies and their underlying infrastructure, as well as the normative framework within which such technologies operate. Currently, this ability is threatened by various challenges, which the Union will need to tackle if it wishes to safeguard and maintain its digital sovereignty. Without the pretention of being exhaustive, in this section I examine five challenges in particular, focusing primarily on the normative dimension of digital sovereignty. Two of these challenges are *external* to the EU, namely the significant role of private power in the public digital sphere (3.1) and malicious interference by foreign actors (3.2). Three of these are *internal*, namely Europe's so-called rule of law crisis (3.3), the risk of poor digital governance (3.4), and a lack of digital literacy (3.5).

3.1 Private power in the digital public sphere

Digital technology and the infrastructure that enables it is almost exclusively in the hands of private companies. A handful of these companies are denoted as 'big tech' or 'tech giants', in light of their dominant or quasi-dominant position in various areas of the tech industry, from artificial intelligence to cloud computing, e-commerce and social networking.⁴¹ Essentially, these private actors make vital choices about the design of the technologies that shape our world in a multitude of ways. The decision of which capabilities, limitations and thus affordances digital technologies have, is riddled with normative implications. Consider, for instance, choices around the order and shape in which information is being shown to us on social media platforms, the content that is filtered on chatbots, the security measures implemented to prevent hacks or data leaks, the datasets on which AI systems are trained, or the ability to engage in end-to-end encrypted communication.

These digital technologies permeate our daily lives not only at an individual level, but also at the level of companies, organizations, public institutions and governments – all of which depend on them. Accordingly, the decisions surrounding their operations and functioning have direct and indirect impacts on society at large, including in ways that we are not yet able to discern. Private actors hence also exercise considerable influence on the public sphere, without, however, being bound by the conditions that typically direct public action, such as the

economy encompasses in particular digital sovereignty in an open manner, respect for fundamental rights, rule of law and democracy, inclusion, accessibility, equality, sustainability, resilience, security, improving quality of life, the availability of services and respect of everyone's rights and aspirations."

⁴¹ See in this regard Damian Tambini and Martin Moore, *Regulating Big Tech: Policy Responses to Digital Dominance*. (Oxford University Press 2022).

requirement to act in the public interest and the need to be able to justify one's actions. In liberal democracies, justifications typically stem from a legal norm that was democratically decided upon and that grants certain competences. However, when it comes to big tech, these decisions with normative implications are being taken by people who work in the interest of the private company that hired them, and without democratic input, even though it affects the public at large. It is hence no surprise that large tech companies are being called 'quasi-sovereign' given the extent of their power.⁴² Interestingly, regulators are also treating the heads of these companies like they would treat heads of state.⁴³

Moreover, when the EU (or any other regulator, for that matter) seeks to deploy its normative digital sovereignty to reign in the power of these companies, these efforts must grapple not only with an inherent asymmetry of information, but also with billions of dollars that the same companies are spending to lobby against restrictive regulations that would undermine their power.⁴⁴ Yet in a less overt manner, these companies also influence EU norms by framing the role of their technologies within a positive narrative of progress, ingenuity, and public interest – through marketing, advertising and other initiatives – which in turn affects public opinion as well as policy initiatives around those technologies.

Furthermore, even when hard law has been passed, it is clear that legislation has its limits. For instance, over five years after its applicability, experience with the GDPR has shown that its safeguards are being underenforced.⁴⁵ And even when investigations are being carried out and infringements of EU competition law or data protection law are found, the amount of the fines that are imposed barely make a difference for companies whose annual revenues are higher than the GDP of several countries. Some have called for the creation of publicly funded alternatives to private tech infrastructures or 'digital public infrastructures'.⁴⁶ The fact that many of these companies are established outside of Europe – and hence might be more difficult to bring under Europe's jurisdiction, despite their effect

⁴² See for instance Broeders, Cristiano and Kaminska (n 25) 2.

⁴³ For instance, a scroll through the Twitter Feed of Sam Altman (CEO of Open.AI) between April and June 2023, shows him having one high-level meeting after another with heads of state, as well as the President of the European Commission, all eagerly posing next to him for a formal picture.

⁴⁴ For instance, most recently, Open AI has been called out for lobbying to water down certain requirements in the upcoming AI Act, as reported by Billy Perrigo, 'Exclusive: OpenAI Lobbied E.U. to Water Down AI Regulation' *Time* (20 June 2023) <<https://time.com/6288245/openai-eu-lobbying-ai-act/>> accessed 22 June 2023. For another example, see reporting around the efforts to weaken the Digital Markets Act before its finalization: Natasha Lomas, 'Report Reveals How Big Tech Lobbied to Weaken EU Rules' *TechCrunch* (22 April 2022) <<https://techcrunch.com/2022/04/22/google-facebook-apple-eu-lobbying-report/>> accessed 22 June 2023.

⁴⁵ See e.g. Estelle Masté, 'Four Years under the GDPR: How to Fix Its Enforcement' (Access Now 2022) <<https://www.accessnow.org/wp-content/uploads/2022/07/GDPR-4-year-report-2022.pdf>> accessed 29 June 2023.

⁴⁶ Ethan Zuckerman, 'The Case for Digital Public Infrastructure' (The Knight First Amendment Institute at Columbia University 2020) <<http://knightcolumbia.org/content/the-case-for-digital-public-infrastructure>> accessed 25 June 2023.

on the European public sphere – has also inspired calls for the establishment of European-based big tech platforms.⁴⁷ Yet regardless of their place of establishment, large private actors in the digital sphere will continue posing a significant challenge to the EU’s ability to autonomously decide over its digital fate. And while the Union has been adopting and negotiating a series of new regulations that aim to tighten its control over such fate, only time will tell how effective these rules will be if private tech companies would prefer not to abide by them.

3.2 Interference by foreign entities

A second challenge is located outside of the Union’s borders, and arises when foreign entities deliberately interfere with Europe’s digital sovereignty. Such interference is neither new nor confined to the EU. Russia has, for instance, been accused of interfering both with the US elections of 2016⁴⁸ and with the UK’s Brexit referendum⁴⁹ which took place that same year, alongside alleged interference attempts in EU member states.⁵⁰ A 2016 Joint Communication by the European Commission and the High Representative of the Union for Foreign Affairs and Security Policy underlined an increase in security challenges – particularly in the form of hybrid threats, described as “*the mixture of coercive and subversive activity, conventional and unconventional methods (i.e. diplomatic, military, economic, technological), which can be used in a coordinated manner by state or non-state actors to achieve specific objectives while remaining below the threshold of formally declared warfare*”.⁵¹

These activities range from blatant cyber-attacks on digital (critical) infrastructure to the initiation of mis- and dis-information campaigns and (social) media manipulation. Evidently, foreign meddling with the EU’s digital and political infrastructure undermines the Union’s digital sovereignty. While securing resilience against such threats is primarily a Member State responsibility, the Union is increasingly seeking to coordinate and consolidate Member State efforts in this

⁴⁷ Jean-François Bobier and others, ‘Can Europe Create Its Own Deep-Tech Giants?’ (Boston Consulting Group 2022) <<https://www.bcg.com/publications/2022/how-can-europe-build-deep-tech-leaders>> accessed 25 June 2023.

⁴⁸ United States Senate, ‘Report of the Select Committee on Intelligence - Russian Active Measures Campaigns and Interference in the 2016 US Election’ (2020) <https://www.intelligence.senate.gov/sites/default/files/documents/Report_Volume1.pdf> accessed 29 June 2023. See also Kathleen Hall Jamieson, *Cyberwar: How Russian Hackers and Trolls Helped Elect a President: What We Don’t, Can’t, and Do Know* (Oxford University Press 2020).

⁴⁹ This accusation is more contentious, and the UK government was criticized for not duly investigating such interference. See also Mark Landler and Stephen Castle, “‘No One’ Protected British Democracy From Russia, U.K. Report Concludes’ *The New York Times* (21 July 2020) <<https://www.nytimes.com/2020/07/21/world/europe/uk-russia-report-brexit-interference.html>> accessed 29 June 2023.

⁵⁰ See Erik Brattberg and Tim Maurer, ‘Five European Experiences With Russian Election Interference’ (Carnegie Endowment for International Peace 2018).

⁵¹ European Commission and High Representative of the Union for Foreign Affairs and Security Policy, ‘Joint Communication to the European Parliament and the Council: Joint Framework on Countering Hybrid Threats - a European Union Response’ (2016) JOIN(2016) 18 final.

context.⁵² Evidently, a vulnerability in one Member State leads to the vulnerability of the EU as a whole. This also led the European Parliament to set up a *Special Committee on Foreign Interference in all Democratic Processes in the European Union, including Disinformation* (INGE) in June 2020. The mandate of this Committee was renewed in March 2022 (ING2), with a final report adopted in June 2023 calling for “a coordinated strategy to increase the EU’s resilience to foreign interference and information manipulation and protect the 2024 European elections.”⁵³

When Russia invaded Ukraine in February 2022, this not only destabilized the global ‘order’ that had been taken for granted by many, but it served as a chilling reminder that these threats are not limited to cyberspace. It also exposed even more clearly Europe’s vulnerability when it comes to physical infrastructures that enable digital technology. If gas pipelines can be sabotaged, are undersea internet cables next?⁵⁴ And if so, how can these be protected? The fact that these cables – on which the internet traffic of virtually the entire world depends – are operated largely by private companies, does not make things easier.

It should be stressed that the threat of foreign interference is certainly not limited to Russia. Malicious cyber activities directed against EU institutions and European companies also emanated from China,⁵⁵ leading the Parliament’s ING2 Special Committee to focus on both Russian and Chinese interference. Furthermore, beyond (cyber)attacks and informational warfare campaigns, foreign countries have other tools to influence the EU’s digital sovereignty. Consider, for instance, European companies’ reliance on Chinese digital technology, which can lead to infrastructural vulnerabilities by facilitating surveillance practices.⁵⁶ Likewise, China’s data regulations can be seen against the same background. As Aynne Kokas persuasively argues, China protects its digital borders by controlling

⁵² See e.g. also High Representative of the Union for Foreign Affairs and Security Policy, ‘Joint Report to the European Parliament and the Council on the Implementation of the Joint Framework on Countering Hybrid Threats - a European Union Response’ (European Commission 2017).

⁵³ European Parliament, ‘Foreign Interference: MEPs Call for Urgent Protection of 2024 European Elections’ (6 January 2023) <<https://www.europarl.europa.eu/news/en/press-room/20230524IPR91908/foreign-interference-meps-call-for-urgent-protection-of-2024-european-elections>> accessed 24 June 2023.

⁵⁴ Mark Scott, ‘Will Russia Attack Undersea Internet Cables Next?’ *POLITICO* (29 September 2022) <<https://www.politico.eu/article/everything-you-need-to-know-about-the-threat-to-undersea-internet-cables/>> accessed 24 June 2023.

⁵⁵ ‘China: Declaration by the High Representative on Behalf of the European Union Urging Chinese Authorities to Take Action against Malicious Cyber Activities Undertaken from Its Territory’ *Council of the European Union* (19 July 2021). See also Stuart Lau, ‘Europe Joins US to Condemn Cyberattacks from China’ *POLITICO* (19 July 2021) <<https://www.politico.eu/article/europe-us-condemnation-china-state-sponsored-cyberattacks/>> accessed 24 June 2023.

⁵⁶ See e.g. Clothilde Goujard, ‘Time to Ban TikTok, EU Lawmakers Tell Governments’ *POLITICO* (2 June 2023) <<https://www.politico.eu/article/time-ban-tiktok-kaspersky-huawei-eu-parliament-lawmakers/>> accessed 25 June 2023; Samuel Stolton and Laurence Cerulus, ‘Belgian Intelligence Puts Huawei on Its Watchlist’ *POLITICO* (28 March 2023) <<https://www.politico.eu/article/belgian-intelligence-huawei-watchlist-espionage-china-eu-nato/>> accessed 25 June 2023.

data and tech infrastructure and meticulously adopted laws “*that enshrine direct government access to corporate data, both domestically and internationally*”, thus facilitating so-called ‘data-trafficking’.⁵⁷ This affects not only individuals but also companies and institutions whose data is being ‘trafficked’. Accordingly, through norm-setting, China is exercising its own digital sovereignty to enlarge its grip over foreign entities. Another example concerns the EU’s criticism of Chinese court rulings that prevent European companies from protecting their intellectual property rights for technological innovations.⁵⁸ These tensions show the inherent link between economic and normative digital sovereignty, as they are largely fueled by the differences in the underlying societal value-systems. Moreover, the global rise in authoritarian tendencies⁵⁹ makes it likely that clashes in value-systems will continue to threaten Europe’s digital sovereignty.

The EU is not entirely defenseless against these threats. The Commission has sought recourse to dispute resolution mechanisms to enforce the rights of EU companies,⁶⁰ and has adopted several initiatives to limit its dependency on foreign technologies. Moreover, various regulations were adopted precisely to better protect the EU’s constituents. The GDPR is, for instance, establishes important safeguards against (individuals’) data trafficking.⁶¹ Furthermore, since October 2020, the Foreign Direct Investment Regulation applies.⁶² Among relevant factors to determine whether a foreign investment merits screening, it lists its effect on inter alia “*critical technologies and dual use items (including artificial intelligence, robotics, semiconductors, cybersecurity, quantum, aerospace, defense, energy storage, and nuclear technologies, nanotechnologies and biotechnologies)*” and on “*access to or the ability to control sensitive information*”.⁶³ In addition, the Commission is currently mulling over a ban

⁵⁷ Aynne Kokas, *Trafficking Data: How China Is Winning the Battle for Digital Sovereignty* (Oxford University Press 2022) 2.

⁵⁸ European Commission, ‘EU Challenges China at the WTO to Defend High-Tech Sector’ (Brussels, 18 February 2022) <https://ec.europa.eu/commission/presscorner/detail/en/ip_22_1103> accessed 25 June 2023..

⁵⁹ Freedom House, ‘Freedom in the World 2022: The Global Expansion of Authoritarian Rule’ (2022) <https://freedomhouse.org/sites/default/files/2022-02/FIW_2022_PDF_Booklet_Digital_Final_Web.pdf> accessed 1 August 2022.

⁶⁰ E.g. before the WTO, see European Commission, ‘EU Challenges China at the WTO to Defend High-Tech Sector’ (n 58).

⁶¹ European Parliament and Council, Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) 2016 1. See also Zuzanna Gulczyńska, ‘A Certain Standard of Protection for International Transfers of Personal Data under the GDPR’ (2021) 11 *International Data Privacy Law* 360.

⁶² Regulation (EU) 2019/452 of the European Parliament and of the Council of 19 March 2019 establishing a framework for the screening of foreign direct investments into the Union (OJ L 79I, 21 March 2019, p 1–14).

⁶³ *Ibid*, Article 4.

on European companies to develop certain sensitive technologies abroad,⁶⁴ just as it is urging Member States to disallow certain foreign companies to participate in their 5G telecoms networks out of security concerns.⁶⁵ Finally, as a response to Russia's invasion of Ukraine, the EU also managed to adopt several heavy sanction packages with the aim of weakening Russia's ability to finance and continue the war. Many of these sanctions explicitly targeted the digital sector, including for instance an imposition of technology export controls.⁶⁶ These examples demonstrate the Union's ability to target foreign countries that threaten its values. At the same time, the effectiveness of those efforts highly depends on the strength of the EU's own (economic) digital sovereignty, the sovereignty of its 'allies', and the way in which other countries and institutions are positioning themselves in this arena.

3.3 Europe's rule of law crisis

The rise in authoritarian tendencies is not confined to countries outside Europe. Over the past decade, several EU Member States have taken actions that weakened human rights protections, undermined the democratic process and eroded the rule of law. From the adoption of laws that blatantly discriminate against people with a different sexual orientation⁶⁷ and judicial reforms that undermine the separation of powers,⁶⁸ to obstructions of civil society organizations and independent media outlets⁶⁹ – collectively, these developments are seen as a 'rule of law crisis' within

⁶⁴ Jakob Hanke and Barbara Moens, 'EU Looks to Ban Companies from Making Sensitive Tech in China' *POLITICO* (20 June 2023) <<https://www.politico.eu/article/eu-ban-companies-make-sensitive-tech-china/>> accessed 25 June 2023.

⁶⁵ Reuters, 'China Slams EU Ban on Huawei, ZTE Demands Equal Treatment' *Reuters* (16 June 2023) <<https://www.reuters.com/business/media-telecom/china-firmly-opposes-huawei-ban-by-some-eu-countries-2023-06-16/>> accessed 25 June 2023.

⁶⁶ See in this regard Michael Adam and Sanne Keijer, 'Sanctions on the Russian Digital Sector: How Effective Are They?' (European Parliamentary Research Service 2023). June 2023 marked the adoption of the 11th sanction package against Russia, see 'Press Release: EU Adopts 11th Package of Sanctions against Russia for Its Continued Illegal War against Ukraine' *European Commission* (Brussels, 23 June 2023) <https://ec.europa.eu/commission/presscorner/detail/en/ip_23_3429> accessed 23 June 2023.

⁶⁷ See in this regard Katinka Linnamäki, 'Not in Front of the Child: Illiberal Familism and the Hungarian Anti-LGBTQ+ "Child Protective Law"' (2022) 10 *Politics and Governance* 16.28/07/2023 12:48:00

⁶⁸ See also Anna Śledzińska-Simon, 'The Rise and Fall of Judicial Self-Government in Poland: On Judicial Reform Reversing Democratic Transition' (2018) 19 *German Law Journal* 1839.

⁶⁹ European Federation of Journalists, 'Hungary Dismantles Media Freedom and Pluralism' (European Federation of Journalists 2019) <<https://europeanjournalists.org/blog/2019/12/03/new-report-hungary-dismantles-media-freedom-and-pluralism/>> accessed 29 June 2023; 'Hungarian Civil Society Raises Alarm over NGO Law Change' (*Euractiv*, 22 April 2021) <<https://www.euractiv.com/section/justice-home-affairs/news/hungarian-civil-society-raises-alarm-over-ngo-law-change/>> accessed 5 September 2022.

the European Union.⁷⁰ The EU's digital sovereignty hence faces an important threat that emanates from within the EU legal order. When the very Member States that make up the Union decide to ignore or violate EU law, or to undermine its primacy, this weakens Europe's ability to maintain its autonomy, also when it comes to its relationship with digital technology.

Crucially, however, the EU seems to treat concerns linked to digital technologies as separate from concerns linked to its rule of law crisis, despite the fact that the former can exacerbate the latter. Scholars have increasingly been mapping the harms that can arise from the ubiquitous use of automated decision-making systems, especially in the public sphere. These encompass, *inter alia*, the risk of bias and discrimination, wide-spread manipulation, mass-surveillance and privacy infringements, and the erosion of the rule of law.⁷¹ Yet at the same time, the EU is (financially) encouraging Member States to digitalize and automate its public services.⁷² And in the upcoming AI act, no difference is made between requirements imposed on high-risk systems used by private companies or by governments. For virtually all systems, no prior verification by an independent entity is needed to assess whether the requirements are met before the system is used. Member States' police forces and migration authorities will hence be able to 'self-assess' the compatibility of their AI systems with the AI Act's requirements.⁷³

Admittedly, an independent national supervisory authority can however carry out *ex post* compliance controls. Moreover, when it comes to the use of facial recognition systems by law enforcement authorities in public places – an extremely sensitive application given its significant impact on several human rights – the suggested additional safeguard consists of 'pre-judicial approval' of the technology's use.⁷⁴ What is, however, the value of a pre-judicial approval if the

⁷⁰ JHH Weiler, 'Europe in Crisis - On Political Messianism, Legitimacy and the Rule of Law' [2012] *Singapore Journal of Legal Studies* 248; Laurent Pech and Kim Lane Scheppele, 'Illiberalism Within: Rule of Law Backsliding in the EU' (2017) 19 *Cambridge Yearbook of European Legal Studies* 3.

⁷¹ Mireille Hildebrandt, 'Algorithmic Regulation and the Rule of Law' (2018) 376 *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 20170355; M Brkan, 'Artificial Intelligence and Democracy' (2019) 2 *Delphi - Interdisciplinary Review of Emerging Technologies* 66; Gabrielle M Johnson, 'Algorithmic Bias: On the Implicit Biases of Social Technology' (2021) 198 *Synthese* 9941; Francesco Ragazzi and others, 'Biometric & Behavioural Mass Surveillance in EU Member States' (Report for the Greens/EFA in the European Parliament 2021) <<http://extranet.greens-efa.eu/public/media/file/1/7297>> accessed 16 February 2022.

⁷² The EU recovery plan, aimed to 'rebuild a post-COVID-19 Europe', explicitly includes a pillar on 'digital transformation' that Member States need to integrate in their recovery and resilience plans to receive a financial contribution.

⁷³ See also Smuha and others (n 37) 37.

⁷⁴ In the Commission's original proposal of the AI Act as published in April 2021, pre-judicial approval would enable law enforcement to rely on the exceptional use of real-time remote biometric identification (see Article 5(3)). In its position on the AI Act adopted in June 2023, the European Parliament removed any exceptions to the use of *real-time* remote biometric identification by law enforcement, but it introduced the safeguard of pre-judicial approval for situations where law enforcement wishes to make use of 'post' remote biometric identification,

executive branch of a Member State continuously seeks to capture the judiciary and undermines its impartiality? What is the value of ex post control by a supervisory authority if its independence cannot be guaranteed? And even if both the judiciary and the supervisory authority would nevertheless carry out their functions in an impartial and independent manner – how much damage can a government cause to human rights, democracy and the rule of law by deploying AI systems at scale in a way that fosters illiberal and authoritarian ends, before its actions are challenged?

These two sets of problems should hence not be seen in isolation from each other. The EU's ability to autonomously decide its digital course risks being undermined if it cannot be ensured that Member States actually abide by its digital policies. Moreover, given the mutual trust that underpins the EU legal system, all Member States are adversely affected when one Member State shies away from its commitments under EU law and is no longer worthy of such trust. In addition, this also affects the EU's external position, as its geopolitical strength – including when it comes to digital matters – hinges on its unity. Accordingly, if the EU wishes its digital policies to succeed, both internally and externally – it is crucial that it gets a better grip on this threat from within.

Some EU institutions have made attempts to tackle these concerns. The European Parliament adopted several resolutions against Hungary⁷⁵ and Poland,⁷⁶ asking the Commission and the Council to take action. The Commission also dragged these Member States before the Court of Justice for rule of law and human rights infringements.⁷⁷ It also recently launched the conditionality mechanism against Hungary, which allows the Union to target breaches of the rule of law that affect or seriously risk affecting the sound financial management of the budget or

namely the analysis of recorded footage of publicly accessible spaces which does not occur in real-time (see Amendment 227).

⁷⁵ See for instance 'European Parliament Resolution of 1 June 2023 on the Breaches of the Rule of Law and Fundamental Rights in Hungary and Frozen EU Funds' (European Parliament 2023) (2023/2691(RSP)) <https://www.europarl.europa.eu/doceo/document/TA-9-2023-0216_EN.html> accessed 28 June 2023.

⁷⁶ See for instance European Parliament, 'European Parliament Resolution of 17 September 2020 on the Proposal for a Council Decision on the Determination of a Clear Risk of a Serious Breach by the Republic of Poland of the Rule of Law (COM(2017)0835 – 2017/0360R(NLE))'; 'European Parliament Resolution of 21 October 2021 on the Rule of Law Crisis in Poland and the Primacy of EU Law' (European Parliament 2021) (2021/2935(RSP)).

⁷⁷ See for instance European Commission, 'Commission Refers Hungary to the Court of Justice of the EU over Violation of LGBTIQ Rights' *European Commission - European Commission* (Brussels, 15 July 2022) <https://ec.europa.eu/commission/presscorner/detail/en/ip_22_2689> accessed 28 June 2023; European Commission, 'The European Commission Decides to Refer Poland to the Court of Justice of the European Union for Violations of EU Law by Its Constitutional Tribunal' *European Commission - European Commission* (Brussels, 15 February 2023) <https://ec.europa.eu/commission/presscorner/detail/en/ip_23_842> accessed 28 June 2023.

the protection of the Union's financial interests.⁷⁸ The Court of Justice issued several judgments⁷⁹ against Member States' actions that are incompatible with the protection of the rule of law, as did the European Court of Human Rights in Strasbourg.⁸⁰ Nevertheless, concerns remain, and no significant improvements are in sight just yet. Furthermore, the values of liberal democracy are inherently fragile in all countries, especially given the increase in populism. Their protection and realization should hence be considered as a continuous work in progress, that should be considered in all domains of EU action. It is hence crucial for the Union to contextualize this internal threat also within the sphere of its digital policies.

3.4 Poor digital governance

A fourth challenge stems from the risk that the EU uses its norm-setting power in an inadequate manner, thereby foregoing to protect its constituents against the adverse effects that digital technologies can generate. If its existing or upcoming digital regulations are deficient, incomplete or ineffective, or if Member States fail to duly implement them, the role of digital technology in European society will evolve in a way that was not desired, or that is incompatible with the EU's values. In essence, through poor digital governance, the EU could also undermine its own digital sovereignty. It is hence not enough for the EU to adopt legislative initiatives in the digital sphere, but it is essential that those initiatives are robust, evidence-based, and capable of steering digital technology in the direction that guarantees their compatibility with human rights, democracy and the rule of law. To concretize the risk of poor digital governance, let me provide an example of the consequences that may arise from poor *AI* governance.

At the time of writing this paper, the European Parliament and the Council are still negotiating the abovementioned AI Act, setting out the conditions under which AI systems can be placed on the market or put into service in the EU. These negotiations take place amidst tensions between those who fear that a strict approach to AI regulation might harm innovation and undermine the benefits it can generate, and those who maintain that strict rules are indispensable to protect EU values against AI's risks. What happens if this balance is not struck in an appropriate manner? Let us imagine that the new AI governance framework errs

⁷⁸ European Commission, 'Proposal for a Council Implementing Decision on Measures for the Protection of the Union Budget against Breaches of the Principles of the Rule of Law in Hungary' (European Commission 2022) COM(2022) 485 final.

⁷⁹ See e.g. Laurent Pech and Dimitry Kochenov, 'Respect for the Rule of Law in the Case Law of the European Court of Justice: A Casebook Overview of Key Judgments since the Portuguese Judges Case' (SIEPS 2021) <<https://www.sieps.se/en/publications/2021/respect-for-the-rule-of-law-in-the-case-law-of-the-european-court-of-justice/>>.

⁸⁰ See in this regard also Elisabeth Steiner, 'The Rule of Law in the Jurisprudence of the European Court of Human Rights' in Werner Schroeder (ed), *Strengthening the Rule of Law in Europe: From a Common Concept to Mechanisms of Implementation* (Hart Publishing 2016); Robert Spano, 'The Rule of Law as the Lodestar of the European Convention on Human Rights: The Strasbourg Court and the Independence of the Judiciary' (2021) 27 European Law Journal 211.

on the side of the former, foregoing any imposition of stringent transparency and accountability rules for AI providers and users, and instead relies on voluntary codes of conduct. This would severely undermine our ability to assess whether the organizations who deploy AI systems abide by EU law, including non-discrimination legislation and data protection regulations, thus diminishing the EU's digital sovereignty.

Moreover, AI systems are also increasingly used to enforce (EU) law, by 'translating' legal rules to code to enable algorithmic regulation (whether by governments, for instance in public administrations, or by companies, for instance in the context of AI-enabled content moderation). Without a proper AI governance mechanism, it will be impossible to verify whether that translation occurred in a manner that is compatible with EU law. Nor will it be possible to ensure that governments aren't somehow relying on those systems to surveil and control their population in a way that is incompatible with EU norms.⁸¹ More generally, the risk is not merely the fact that an array of intrusive AI applications – from facial recognition systems to predictive policing and manipulative chatbots – will be rolled-out in all domains of our lives and affect human rights. The main danger is that this roll-out proves irreversible as AI systems become an inextricable part of existing infrastructures. High investments, path dependencies, and the gradual habituation of these systems across the population will lead to their societal entrenchment, even if undermining our individual and collective autonomy, and thus also our sovereignty.⁸²

The EU legislator currently has a unique opportunity to decide over Europe's AI future – one that might be difficult to revisit once the AI Act is adopted. Despite uncertainties about the technology's evolution, the information asymmetries between AI developers and regulators, and the lack of a proper understanding of the technology's (long term) impact on society, the EU will need to take a stance on AI's governance and ensure that the normative framework it puts in place aligns with its goals and values. This is not an easy task, especially in light of the Union's desire to also foster the technology's development within its borders and reduce dependencies on foreign technology providers. Yet given the rapid and widespread implementation of ever more powerful AI systems, there is little time for regulatory experimentation that can help ensure a regulatory race to the top, since not (yet) regulating is also a choice, and one that the EU can ill afford.⁸³ It will hence need to rely on its vast experience with (other technology)

⁸¹ See in this regard also Marvin van Bekkum and Frederik Zuiderveen Borgesius, 'Digital Welfare Fraud Detection and the Dutch SyRI Judgment' [2021] *European Journal of Social Security* 13882627211031256.

⁸² See in this regard Karen Yeung, 'Why Worry about Decision-Making by Machine?' in Karen Yeung and Martin Lodge (eds), *Algorithmic Regulation* (Oxford University Press 2019); Karen Yeung, 'Responsibility and AI - A Study of the Implications of Advanced Digital Technologies (Including AI Systems) for the Concept of Responsibility within a Human Rights Framework' (Council of Europe 2019) DGI(2019)05.

⁸³ Smuha, 'From a "Race to AI" to a "Race to AI Regulation"' (n 38) 81.

regulation to ensure an approach that is both pungent and flexible in protecting its digital fate.

3.5 Lack of digital literacy

The final challenge I wish to highlight concerns the need for an appropriate level of digital literacy within the European Union. After all, to exercise digital sovereignty, one must have an understanding of what digital technologies are, how they function, what their capabilities and limitations are, and how they affect society. This holds true for regulators at national and EU level that adopt digital policies, but it is equally important for citizens, who must be able to express their informed opinion, within a democratic process, of the type of digital future they envisage for society. Studies have for instance found a positive correlation between digital literacy and online political engagement.⁸⁴ In essence, digital literacy can be seen as a precondition for digital sovereignty.

According to Eurostat, in 2021 only 54% of people in the EU aged 16 to 74 had basic overall digital skills.⁸⁵ Digital skills in this context refers to information and data literacy skills, communication and collaboration skills, digital content creation skills, safety skills and problem-solving skills.⁸⁶ To increase this number, the Commission launched the Digital Education Action Plan, aimed at improving “*digital literacy, skills and capacity at all levels of education and training and for all levels of digital skills*”.⁸⁷ While education primarily remains a Member State competence, especially as regards the content of educational programs and their organization, the Commission has encouraged coordinated actions at the level of the EU, from the exchange of best practices to supporting cooperation in the educational realm.

Importantly, digital literacy includes, but goes beyond, digital skills. It also implies the ability to critically think about and engage with digital technologies, and to understand their (adverse) effects on individuals and society. In the context of AI literacy, for instance, the High-Level Expert Group on AI noted that this covers “*fostering the understanding of AI systems more generally (including a basic understanding of*

⁸⁴ Joseph Kahne, Nam-Jin Lee and Jessica Timpany Feezell, ‘Digital Media Literacy Education and Online Civic and Political Participation’ (2012) 6 *International Journal of Communication* 1; See also Joseph Kahne and Benjamin Bowyer, ‘Can Media Literacy Education Increase Digital Engagement in Politics?’ (2019) 44 *Learning, Media and Technology* 211; Yunrong Li and Gang Li, ‘The Impacts of Digital Literacy on Citizen Civic Engagement—Evidence from China’ (2022) 3 *Digital Government: Research and Practice*.

⁸⁵ ‘How Many Citizens Had Basic Digital Skills in 2021?’ (Eurostat 2022) <<https://ec.europa.eu/eurostat/web/products-eurostat-news/-/ddn-20220330-1>> accessed 26 June 2023.

⁸⁶ *Ibid.* To be classified as having ‘basic overall digital skills’, a person must be able to carry out at least one activity related to each of those areas.

⁸⁷ European Commission, ‘Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions: Digital Education Action Plan 2021-2027 - Resetting Education and Training for the Digital Age’ (2020) SWD(2020) 209 final. The first such plan was adopted in January 2018 as part of the initiative to establish a European Education Area.

*machine learning and reasoning), but also raising awareness of data protection rights, an understanding of how (personal) data can be used, the implications of digital tracking, and the importance of issues such as fairness, explainability, transparency, robustness of AI systems, and knowledge of these topics.”*⁸⁸ The lack of such understanding stands in the way of participating in public debates, which are increasingly taking place in the digital sphere. Yet it also diminishes people’s ability to realize the impact that digital technology might have on their lives. This, in turn, diminishes their ability to formulate their views on which impact they deem compatible with societal values, and to urge their political representatives to take proper action.

Finally, ensuring digital literacy is not only crucial to *adopt* but also to *enforce* the digital regulations through which the EU is exerting and protecting its digital sovereignty. The future AI Act, for instance, hinges on the development of standards through which compliance with its requirements can be assessed and compared, from AI fairness to transparency.⁸⁹ Developing such standards requires an understanding of both the technical aspects of the technology and its effects on individuals and society, including for instance how it might affect fundamental rights. There is hence a need for people with multidisciplinary skills to ensure that the AI Act’s aspired safeguards are effectively implemented.⁹⁰ Similarly, Member States will need to appoint a supervisory authority that can verify compliance with the AI Act’s requirements.⁹¹ Yet besides financial resources, such authorities will only be able to properly carry out their tasks if they also have the necessary *human* resources with sufficient competences and skills. To secure their digital sovereignty also over the longer term, EU institutions and Member States alike will hence need to step up their efforts and augment Europe’s digital literacy sooner rather than later.

4 Conclusions

The European Union has put digital sovereignty increasingly high on its agenda. The ubiquity of digital technologies in all domains of society has prompted concerns around the level of control we still have over our relationship therewith, and over their effects on individuals, companies and institutions – for better and for worse. Strengthening digital sovereignty is thus seen as a means to secure such control, and in particular to maintain autonomy over Europe’s digital ‘destiny’. The

⁸⁸ High-Level Expert Group on AI, ‘Policy and Investment Recommendations for Trustworthy AI’ (European Commission 2019) 10 <<https://digital-strategy.ec.europa.eu/en/library/policy-and-investment-recommendations-trustworthy-artificial-intelligence>>.

⁸⁹ According to Recital 61 of the Commission’s proposal for an AI Act, “*standardisation should play a key role to provide technical solutions to providers to ensure compliance with this Regulation. Compliance with harmonised standards as defined in Regulation (EU) No 1025/2012 of the European Parliament and of the Council 54 should be a means for providers to demonstrate conformity with the requirements of this Regulation.*”

⁹⁰ See in this regard also the example in Francesco Sovrano and others, ‘Metrics, Explainability and the European AI Act Proposal’ (2022) 5 J 126.

⁹¹ See Title VI of the proposed AI Act.

scale at which these technologies function and our increased dependency thereon has given the matter a sense of urgency.⁹² As the Commission and the High Representative noted in their most recent Communication on the matter, “*the global pandemic, Russia’s illegal and unprovoked war in Ukraine, hostile economic actions, cyber and infrastructure attacks, foreign interference and disinformation and a global increase in geopolitical tensions have exposed risks and vulnerabilities in our societies, economies and companies that did not exist only a few shorts year ago.*”⁹³

In this paper, I sought to unpack the notion of digital sovereignty in the EU through an exploration of its use in European discourse. While digital sovereignty is elastic and knows various dimensions, a dual thread seems to run across many of its conceptions, namely the ability to control the production factors of digital technologies and their underlying infrastructure (economic digital sovereignty) and the ability to control the normative framework within which such technologies operate (normative digital sovereignty). Focusing primarily on the latter, I subsequently examined five challenges that the Union must tackle to secure its sovereignty ambitions in the digital realm. Yet the economic and the normative are intrinsically entwined, and the question remains to which extent the EU can impose its values on digital technology (providers) without also ramping up its global digital market share.

Without a doubt, the regulatory choices and investments that the Union will adopt during this ‘digital decade’ will be determinative of Europe’s digital fate for decades to come. A delicate balance will need to be struck between the potential for innovation and protective regulation, between economic gains and individual rights, and between international cooperation and strategic independence. There is no single way to govern digital technologies, and only time will tell if the course adopted by the EU will generate the effects it intended. Yet as long as it bears in mind the constitutional boundaries of its sovereignty – namely the need to protect human rights, democracy and the rule of law – there is hope that Europe can harness technology in a way that not only safeguards but even fosters the very values that ground its legal order.

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⁹² See also Madiega Tambiama, ‘Digital Sovereignty for Europe’ (European Parliamentary Research Service 2020) EPRS Ideas Paper.

⁹³ European Commission and High Representative of the Union for Foreign Affairs and Security Policy, ‘Joint Communication to the European Parliament, the European Council and the Council on “European Economic Security Strategy”’ (European Commission 2023) JOIN(2023) 20 final.